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SUBJECT CODE NO: - RNP-8005
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - III
Examination November / December 2025 (To be held in February-2026)
Transformer and DC Machine

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

1. Section A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Section B: Q.2 to Q.7.

Section A

Q1 Solve any 10 following questions

20

- 1) Commutator converts -----
 - a) DC to AC
 - b) AC to DC
 - c) High voltage to low voltage
 - d) Low current to high current
- 2) Armature winding is placed on.....
 - a) Stator
 - b) Rotor
 - c) Shaft
 - d) Yoke
- 3) The function of tap changer is to -----
 - a) Increase frequency
 - b) Control voltage
 - c) Reduce power factor
 - d) Increase current
- 4) In an ideal transformer, input power is equal to.....
 - a) Zero
 - b) Output power
 - c) Copper loss
 - d) Core loss
- 5) Efficiency of a transformer is maximum when -----
 - a) Core loss = Copper loss
 - b) Core loss > Copper loss
 - c) Copper loss = 0-meter
 - d) Total loss is minimum
- 6) A DC machine works on the principle of -----
 - a) Lenz's law
 - b) Faraday's law of electromagnetic induction
 - c) Coulomb's law
 - d) Gauss law

- 7) Back EMF acts:.....
 - a) Along applied voltage
 - b) Opposite supply voltage
 - c) Perpendicular to current
 - d) Only at no-load
- 8) The starting torque of DC series motor is -----
 - a) Low b) Medium c) Very high d) Zero
- 9) Which DC generator has the poorest voltage regulation?
 - a) Shunt generator b) Series generator
 - c) Compound generator d) Differential compound generator
- 10) The main function of a yoke in a DC machine is -----
 - a) Provide mechanical support b) Complete the magnetic path
 - c) Protect the machine d) All of the above
- 11) The EMF generated in a DC generator is:
 - a) AC b) DC c) Both AC and DC d) None
- 12) High starting current in DC motor is due to:
 - a) High back EMF b) Low armature resistance
 - c) High field flux d) Mechanical load

SECTION B

Solve any four from the following

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|----|---|----|
| Q2 | a) Explain Construction & Working principle of D.C generator | 05 |
| | b) Explain Construction & Working principle of 1Phase Transformer | 05 |
| Q3 | a) Explain: Speed control of DC Shunt motor | 05 |
| | b) With the help of neat circuit diagram explain how O.C. & S.C. test is conducted on single-phase transformer. | 05 |
| Q4 | a) Why is core of transformer laminated? Explain? | 05 |
| | b) What are the different types of losses in transformer & explain it? | 05 |
| Q5 | a) Write a Short Note on BLDC Motor | 05 |
| | b) Explain: Armature Reaction in DC Machines. | 05 |

- Q6 a) A 4 pole DC Shunt generator with lap connected armature has field and armature resistance of 80 ohm and 0.1 ohm respectively. It supplies power to 50 lamps, rated to 100 volts, 60 watt each. Calculate total armature current and generated emf, by allowing a brush drop of 2V per brush. **05**
- b) Explain: Power Flow Diagram of DC Machine **05**
- Q7 a) Explain: Losses in DC Machines. **05**
- b) Write a Short Note on Three-point starter. **05**